O-M-232e
July 30, 1968
SUPERSEDING
Fed. Spec. O-M-232d
June 16, 1960

FEDERAL SPECIFICATION

METHANOL (METHYL ALCOHOL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers methanol of synthetic grades and a denaturing grade.

1.2 Classification. Methanol shall be of the following grades, as specified (see 6.2):

Grade A - Synthetic, 99.85 percent by weight (solvent use).
Grade AA - Synthetic, 99.85 percent by weight (hydrogen-carbon dioxide generation use).
Grade C - Wood alcohol (denaturing grade).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-G-10157 - Generator and Charging Plant, Hydrogen and Carbon Dioxide, Semitrailer-Mounted.

Military Standards:

MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.
MIL-STD-290 - Packaging, Packing, and Marking of Petroleum and Related Products.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Treasury Department:

Internal Revenue Service Regulations, 26-CFR-212 - Formulas for Denatured Alcohol.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should cite "26-CFR-212".)

FSC 6810

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American Society for Testing and Materials (ASTM) Standards:

D 1078 - Distillation Range of Volatile Organic Liquids.
D 1209 - Color of Clear Liquids (Platinum-Cobalt Scale).
D 1296 - Residual Odor of Lacquer Solvents and Diluents.
D 1353 - Nonvolatile Matter in Volatile Solvents for Use in Paint, Varnish, Lacquer, and Related Products.
D 1363 - Permanganate Time of Acetone and Methanol.
D 1364 - Water in Volatile Solvents (Fischer Reagent Titration Method).
D 1612 - Acetone in Methyl Alcohol (Methanol).
D 1613 - Acidity in Volatile Solvents and Chemical Intermediates Used in Paint, Varnish, Lacquer, and Related Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Grades A and AA. Grades A and AA shall conform to table I.

TABLE I. Requirement for grade A and grade AA

Characteristics	Grade A		Grade AA	
	Requirement	Test para.	Requirement	Test para.
Acetone and aldehydes, percent max.	0.003	4.4.2	0.003	4.4.2
Acetone, percent max.	-----	-----	0.001	4.4.3
Ethanol, percent max.	-----	-----	0.001	4.4.3
Acidity (as Acetic Acid) percent max.	0.003	4.4.4	0.003	4.4.4
Alkalinity, percent max. as NH_3	0.003	4.4.5	0.003	4.4.5
Appearance	Clear and colorless	4.4.6	Clear and colorless	4.4.6
Carbonizable substances	No discoloration	4.4.7	No discoloration	4.4.7
Color	Not darker than color standard no. 5 of ASTM platinum-cobalt scale	4.4.8	Not darker than color standard no. 5 of ASTM platinum-cobalt scale	4.4.8
Distillation range	Not more than 1° C. and shall include 64.6° C. ± .10°C. at 760 mm.	4.4.9	Not more than 1° C. and shall include 64.6° ± .10° C. at 760 mm.	4.4.9
Hydrocarbons	No cloudiness or opalescence	4.4.10	No cloudiness or opalescence	4.4.10
Specific gravity, max.	0.7928 at 20°/20°C.	4.4.11	0.7928 at 20°/20° C.	4.4.11
Percent methanol by weight, min.	.99.85	4.4.11	99.85	4.4.11
Nonvolatile content, percent max.	0.0010	4.4.12	0.0010	4.4.12
Odor	Characteristic, nonresidual	4.4.13	Characteristic, nonresidual	4.4.13
Permanganate test	No discharge of color in 30 minutes	4.4.14	No discharge of color in 30 minutes	4.4.14
Water, percent max.	0.15	4.4.15	0.10	4.4.15

3.2 Grade C. Grade C wood alcohol (denaturing grade) shall comply with Internal Revenue Service Regulations SDA1 of 26-CFR-212, Formulas for Denatured Alcohols.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of all methanol of one grade offered for delivery at one time.

4.3.2 Drums and small containers. From each lot, two containers shall be selected at random. From each of the two containers, one-quart specimens shall be taken and placed in separate, clean, dry, glass containers.

4.3.3 Tank cars. Samples shall be taken in accordance with method 8001, Fed. Test Method Std. No. 791.

4.3.4 Lot acceptance. Samples selected as specified in 4.3.2 or 4.3.3 shall be tested as specified in 4.4. If any specimen fails any test the lot which the specimen represents shall be rejected.

4.4 Quality conformance inspection.

4.4.1 Chemical reagents. Wherever chemical reagents are specified in the following tests, analytical reagent grade shall be used.

4.4.2 Acetone and aldehydes. Determine the acetone and aldehyde content in accordance with ASTM D 1612. Nonconformance to table I shall constitute failure of this test.

4.4.3 Acetone and ethanol (grade AA only). Determine the acetone and ethanol content by the elution method of gas chromatography using internal standards.

4.4.3.1 Apparatus and operating conditions. A conventional chromatographic technique utilizing an instrument at least equal in performance to the highest sensitivity range of the Perkin Elmer Vapor Phase Fractometer Model 154 shall be used. A two meter elution column using tetraethylene glycol dimethyl ether (TEEME) or equal as the liquid phase shall be used under the following operating conditions:

Temperature	plus 55° C.
Pressure	adjust to produce helium flow rate of 80 plus or minus 10 ml.
Sample size	0.20 ml.

4.4.3.2 Quantitative analysis.

(a) Internal standards for quantitative analysis curves shall be run on standard specimens prepared by adding pure ethanol or acetone in 0.001 percent by volume increments. The area under the peak is equivalent to the concentration of ethanol or acetone in the specimen, and the peak height is proportional to the concentration. Using a simple ratio, the concentration of ethanol or acetone can be calculated.

(b) The area under the ethanol or acetone curve of the unknown shall be no more than one half that under the ethanol and acetone curve of the 0.001 percent ethanol or methanol standard specimen. Nonconformance to table I shall constitute failure of this test.

4.4.4 Acidity. Determine the acidity in accordance with ASTM D 1613. Nonconformance to table I shall constitute failure of this test.

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4.4.5 Alkalinity.

(1) Introduce 50 ml. of the sample into a 125 ml. Erlenmeyer flask. Add 3 to 5 drops of phenolphthalein indicator and mix thoroughly. Examine visually against a white background for any evidence of pink color. If pink color is detected, qualitative analysis shall be performed in accordance with (2).

(2) To 25 ml. of water, add 1 drop of 0.1 percent methyl red indicator solution (60 percent alcohol) and 0.01N acid or base until the solution is just pink. Add 25 ml. of specimen, mix well, then titrate with 0.01N sulfuric acid until the pink color is restored. Calculate the percent NH_3 by weight as follows:

$$\frac{(\text{ml. of } H_2SO_4) \times (\text{Normality of } H_2SO_4) \times 1.703}{(\text{ml. of specimen}) \times 0.79} = \text{Percent alkalinity as } NH_3$$

Nonconformance to table I shall constitute failure of this test.

4.4.6 Appearance. The specimen shall be thoroughly mixed and shall be examined by transmitted light for color, clarity, suspended matter, or sediment. This is most readily done by pouring some of the specimen into a test tube or Nessler tube. Nonconformance to table I shall constitute failure of this test.

4.4.7 Carbonizable substances. Pipette 5 ml. of the sample into a 20 x 150 mm. test tube. Slowly and carefully add 5 ml. of concentrated sulfuric acid and swirl the tube gently during the addition. DO NOT COOL THE MIXTURE. Allow the tube and contents to stand for 5 minutes at room temperature. Examine the contents visually for any evidence of discoloration. Nonconformance to table I shall constitute failure of this test.

4.4.8 Color. Determine color in accordance with ASTM D 1209. Nonconformance to table I shall constitute failure of this test.

4.4.9 Distillation range. Determine the distillation range in accordance with ASTM D 1078, with the following exceptions:

(a) Measure the 100 ml. of specimen and distillate at room temperature (without water bath).
(b) Determine temperature using JL-4 type thermometer having a range of 45° to 75° C. in increments of 0.1° C.

Nonconformance to table I shall constitute failure of this test.

4.4.10 Hydrocarbons. Dilute 15 ml. of the specimen to 45 ml. with distilled water in a Nessler tube. Visually examine for cloudiness or opalescence during 30 minutes of standing. Nonconformance to table I shall constitute failure of this test.

4.4.11 Specific gravity (methanol content). Determine the specific gravity with a pycnometer of about 50 ml. capacity or by any other method accurate to the fourth decimal place. In the temperature range 15° to 25° C., the specific gravity changes 0.00092 per degree centigrade. Determine the methanol content from the specific gravity based on the following:

Percent by weight	Specific gravity at 20°/20° C.
99.7	0.79325
99.8	0.79297
99.85	0.79283
99.9	0.79269
100.0	0.79239

Nonconformance to table I shall constitute failure of this test.

4.4.12 Nonvolatile content. Determine nonvolatile content in accordance with ASTM D 1353. Nonconformance to table I shall constitute failure of this test.

4.4.13 Odor. Determine odor in accordance with ASTM D 1296. Nonconformance to table I shall constitute failure of this test.

4.4.14 Permanganate test. Determine permanganate test in accordance with ASTM D 1363. Nonconformance to table I shall constitute failure of this test.

4.4.15 Water content. Determine water content in accordance with ASTM D 1364. Nonconformance to table I shall constitute failure of this test.

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DANGER! FLAMMABLE
VAPOR HARMFUL
MAY BE FATAL OR CAUSE BLINDNESS, IF SWALLOWED
CANNOT BE MADE NONPOISONOUS

Keep away from heat and open flame.
Keep container closed.
Avoid prolonged or repeated breathing of vapor.

Use only with adequate ventilation.



POISON



First Aid — Antidote

If swallowed, induce vomiting by any available means - gagging - soapy water - salt solutions - and CALL A PHYSICIAN. Have patient lie down and keep warm. Cover eyes to exclude light.

FIGURE 1.—Poison label for methanol

4.5 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-STD-290.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Methanol, in quantities as specified (see 6.2), shall be packaged, packed, and marked in accordance with the applicable requirements for packaging of solvents as specified in MIL-STD-290 and the special marking requirements of 5.2. The level of packaging shall be level A or C and the level of packing shall be level A, B, or C as specified (see 6.2), with the following additional requirement: Unless otherwise specified (see 6.2), plan B coating with side seam stripping for one-quart cans and one-gallon cans shall be required in accordance with referenced specification. When specified (see 6.2), pails shall be palletized in accordance with MIL-STD-147.

5.2 Special marking. Each container and palletized load shall be labeled as shown in figure 1 in addition to that required in MIL-STD-290. The label shall be waterproofed.

6. NOTES

6.1 Intended use.

6.1.1 Grade A. Grade A methanol is intended to be used as industrial solvent.

6.1.2 Grade AA. Grade AA methanol is intended to be used for the generation of hydrogen and carbon dioxide in a generator conforming to MIL-G-10157.

6.1.3 Grade C. Grade C wood alcohol is a product of destructive distillation of wood and is intended to be used as a denaturant.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grade required (see 1.2).
(c) Unit quantity (see 5.1).
(d) Selection of applicable level of packaging and packing required (see 5.1).
(e) Size of container required (see 5.1).
(f) When plan B coating and side seam stripping is not required (see 5.1).
(g) When palletized loads are required (see 5.1).

6.3 Grade B deletion. Grade B technical, 99.0 percent by weight (solvent use) has been deleted from this specification because it is no longer used.

MILITARY INTEREST:

Custodians:

Army - ME
Air Force - 68

Review activities:

Army - GL, MD, MU
Air Force - 68, 85

User activities:

Army - AT, EL, MR
Navy - OS, AS

Preparing activity:

Army - ME

CIVIL AGENCY INTEREST:

GSA

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